NLP-Based Repeated Question Analyzer and Cross-Year Topic Similarity Retrieval System for Multi-Department Academic Exam Papers Using TF-IDF and Cosine Similarity

A Cross-Year Question Paper Analysis Pipeline for Higher-Education Institutions

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Abstract

This project presents a domain-agnostic Natural Language Processing (NLP) pipeline that automates the detection of repeated and topically related questions across previous-year academic examination papers (PYQs). Combining Optical Character Recognition (OCR), rule-based text cleaning, TF-IDF vectorization, and cosine similarity retrieval, the system was benchmarked on **111 questions across 5 courses**, achieving a peak precision of **1.000** and a balanced F1-score of **0.435**, validating its viability as a scalable academic revision and audit tool.

1. Title

NLP-Based Repeated Question Analyzer and Cross-Year Topic Similarity Retrieval System for Multi-Department Academic Exam Papers Using TF-IDF and Cosine Similarity

2. Problem Statement

Previous-year question papers (PYQs) are a cornerstone revision and diagnostic resource in higher education, yet manually comparing years of papers across departments is slow, inconsistent, and prone to oversight.

Key challenges:

Challenge	Description
Scan-only legacy papers	Most historical papers exist only as scanned PDFs with no digital text layer.
Scenario rephrasing	Outcome-Based Education (OBE) formats often wrap identical core concepts in new narrative case studies, obscuring repetition.
Lexical variation	Recurring concepts reappear with different terminology, parameters, and numerical values year to year.

There is a clear need for an automated, department-agnostic NLP pipeline that ingests scanned exam PDFs, cleans and segments them into individual questions, and retrieves topically related cross-year question pairs.

3. Objectives

1. **Department-Agnostic OCR & Extraction** — Build a cached document pipeline using PyMuPDF and Tesseract OCR to digitize scanned exam PDFs from any academic stream.
2. **Deterministic Layout Cleaning & Segmentation** — Use regex filtering and a state-machine segmenter to strip boilerplate and isolate individual questions.
3. **NLP Preprocessing & Feature Engineering** — Apply tokenization, stopword filtering, lemmatization, and per-course N-gram TF-IDF vectorization.
4. **Cross-Year Similarity Retrieval** — Compute pairwise cosine similarity across exam cycles to detect recurring questions and topics.
5. **Empirical Benchmark Evaluation** — Measure Precision, Recall, and F1-score against human-annotated ground truth across multiple thresholds.
6. **Interactive Visualization Dashboard** — Deploy a Streamlit app for adjustable similarity thresholds and side-by-side question comparison.

4. Dataset

4.1 Overview

A testbed corpus of **10 previous-year examination papers** spanning **5 representative courses** (2 exam cycles each) was used for evaluation. The pipeline architecture is fully department-agnostic and extensible to any discipline.

4.2 Source

Attribute	Detail
Institution	Marian College Kuttikkanam (Autonomous)
Exam Types	Summative End Assessment (SEA II), In-Semester Assessment (ISA), Regular Degree Examinations
Format	Scanned multi-page PDFs (36 pages total), no selectable text layer

4.3 Size

- **Question papers:** 10 PDFs
- **Total pages:** 36 (100% OCR-processed)
- **Extracted questions:** 111
- **Ground-truth duplicate/same-topic pairs:** 39

4.4 Record Attributes

Field	Description
subject_id	Course/subject identifier (e.g., operating_systems)
year	Academic year or exam cycle
q_no	Extracted question number
marks	Assigned mark weight
raw_text	Unnormalized question text
clean_text	Preprocessed, lemmatized text used for vectorization

4.5 Course Breakdown (Benchmark Testbed)

Subject / Course	Year A	Year B	Q's (A / B)	Total Q's	Gold Pairs
Data Structures	2025	2026	16 / 12	28	8
Digital Fundamentals	2024 (SEA II)	2025 (SEA II)	12 / 11	23	7
Discrete Mathematics	2024 (ISA)	2025 (SEA II)	2 / 11	13	4
Fundamentals of Programming (C++)	2024 (SEA II)	2025 (SEA II)	10 / 10	20	8
Operating Systems	2024	2026	16 / 11	27	12
Total	—	—	**56 / 55**	**111**	**39**

4.6 Data Preparation

1. **Hybrid Extraction** — 300 DPI page rendering (PyMuPDF) + Tesseract OCR, with persistent caching in processed/raw/.

2. **Boilerplate Stripping** — 13+ compiled regex rules remove headers, instructions, and Course Outcome tables while preserving structural delimiters (BUNCH, OR).
3. **Grammar-Aware Segmentation** — A boundary state machine detects section transitions and (COx: N Marks) tags, tolerant of OCR misreads (e.g., 1A vs lA).
4. **Override & Verification Pass** — A version-controlled override dictionary (processed/segment_overrides/) resolves 16 edge-case OCR fragments spanning page breaks.

5. Methodology

Scanned Exam PDFs (Any Department)

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1. Hybrid OCR Extraction (PyMuPDF + Tesseract, 300 DPI)

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2. Boilerplate & Header Removal (Regex Blocklist)

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3. Question Segmentation (State Machine + Override Pass)

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4. NLP Preprocessing (Tokenization, Stopwords, Lemmatization)

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5. Per-Course TF-IDF Vectorization (Unigrams + Bigrams)

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6. Cosine Similarity Computation ($n_A \times n_B$ Matrix)

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7. Dynamic Thresholding & Retrieval —► Streamlit Interactive UI

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8. Evaluation Against Gold Annotations (Precision, Recall, F1)

Stage details:

1. **Text Extraction** — PyMuPDF checks for embedded text; if absent (<20 characters), the page is rendered at 300 DPI and passed through Tesseract OCR v5.
2. **Boilerplate Removal** — Regex rules discard institutional headers, codes, and general instructions to prevent false similarity inflation.
3. **Question Segmentation** — Structural markers (BUNCH I–V, OR, (COx: N Marks)) define question boundaries, producing structured JSON records.
4. **NLP Preprocessing** — Lowercasing, punctuation/noise removal, stopword filtering, and WordNet lemmatization (e.g., *calculating*, *calculation* → *calculate*).
5. **TF-IDF Vectorization** — Fitted per course on unigrams and bigrams to preserve subject-specific terminology without cross-course dilution.
6. **Cosine Similarity Computation** — For every Year A question *u* and Year B question *v*:
7. **Dynamic Threshold Retrieval** — Filters pairs exceeding a user-selected threshold, $\tau \in [0.05, 0.60]$.
8. **Evaluation** — Retrieved pairs are scored against gold annotations for Precision, Recall, and F1.

6. Implementation

Language: Python 3.10+

Category	Libraries
PDF & OCR	pymupdf (fitz), pytesseract, pdf2image, Pillow
NLP	nltk (tokenizers, stopwords, WordNet lemmatizer)
Feature Extraction & Similarity	scikit-learn (TfidfVectorizer, cosine_similarity), numpy

Category	Libraries
Data Handling	pandas, json, re, pathlib
Interactive UI	streamlit

NLP Techniques: OCR (300 DPI), regex-based normalization, NLTK stopword filtering, WordNet lemmatization, and combined unigram + bigram modeling to capture multi-word concepts (*"binary search"*, *"page replacement"*, *"linear equation"*).

Core Algorithm:

$$TF\text{-}IDF(t,d,D)=TF(t,d)\times(\ln(1+DF(t,D)1+|D|)+1)$$

followed by pairwise cosine similarity across sparse document matrices.

Implementation notes:

- **Per-course fitting** ensures discriminative weighting for subject-specific vocabulary.
- **Intermediate caching** under processed/ enables sub-second UI response times.
- **Modular design** — adding a new subject requires only a new PDF and a papers.json entry.

7. Results

7.1 Overall Benchmark Across Thresholds

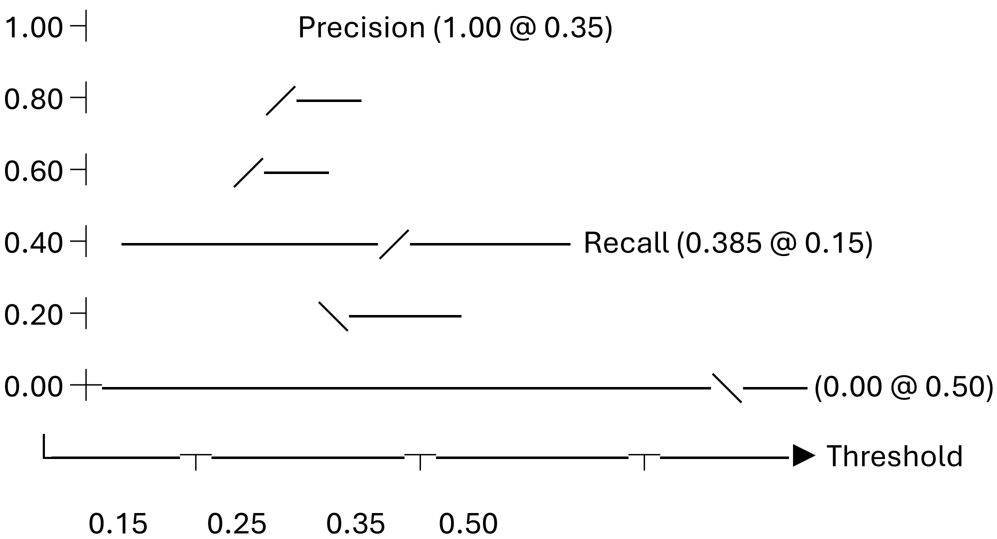
Threshold (τ)	Precision	Recall	F1-Score	Retrieved	TP	FP	FN
0.15 *(default)*	0.500	**0.385**	**0.435**	30	15	15	24
0.25	0.750	0.154	0.255	8	6	2	33
0.35	**1.000**	0.051	0.098	2	2	0	37
0.50	0.000	0.000	0.000	0	0	0	39

7.2 Per-Subject Performance (τ = 0.15)

Subject	Gold Pairs	Retrieved	TP	FP	Precision	Recall	F1
Operating Systems	12	13	8	5	**0.615**	**0.667**	**0.640**
Digital Fundamentals	7	4	3	1	0.750	0.429	0.545
Data Structures	8	11	3	8	0.273	0.375	0.316
C++ Programming	8	2	1	1	0.500	0.125	0.200
Discrete Mathematics	4	0	0	0	0.000	0.000	0.000

7.3 Precision–Recall Trade-off

Precision & Recall Trade-off Across Similarity Thresholds



7.4 Case Studies

✅ **True Positive Match (score 0.41)**

- *Year A:* Page reference string question asking for FIFO/LRU page faults with four frames.
- *Year B:* Equivalent question using a 4-frame FIFO page replacement scenario.
- **Why it matched:** Strong lexical overlap — "page replacement algorithm," "page faults," "reference string."

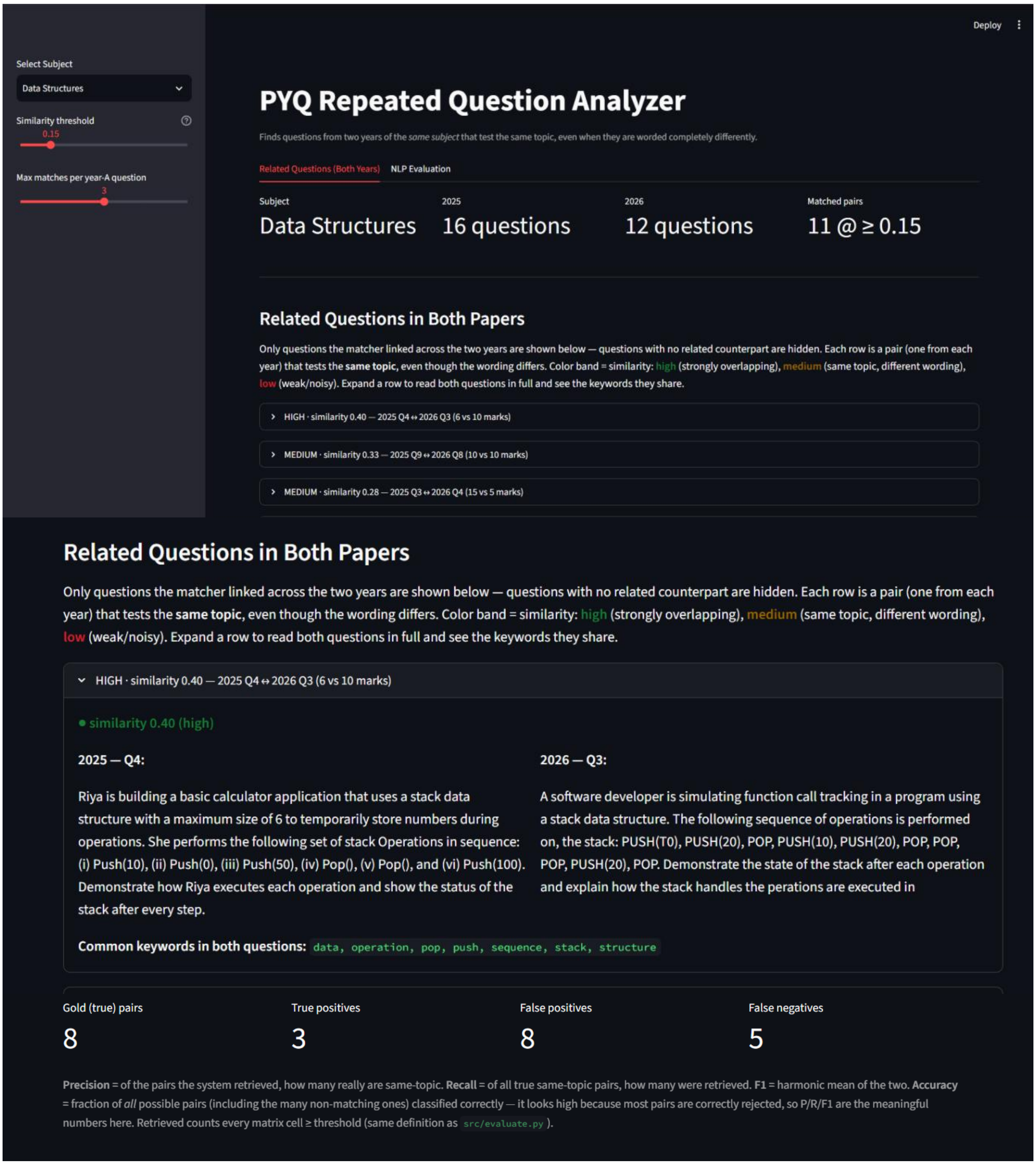
⚠️ **False Positive (τ = 0.25)**

- *Year A:* Deletion of a node in a doubly linked list.
- *Year B:* Insertion of a node in a circular linked list.

- **Why it matched:** Shared vocabulary ("linked list," "node," "position") despite different operations.

❌ **False Negative (narrative variance)**

- *Year A:* BCD addition framed around a digital watch.
- *Year B:* BCD addition framed around a digital weighing scale.
- **Why it was missed:** Divergent narrative terms diluted the shared concept, pulling the similarity score (~0.14) just below threshold.



Metrics across thresholds

Same table as `processed/eval/results.md` — where precision and recall cross is the sweet spot.

threshold	precision	recall	F1	accuracy	retrieved	TP	FP	FN	gold
0.15	0.273	0.375	0.316	0.932	11	3	8	5	8
0.25	0.667	0.25	0.364	0.964	3	2	1	6	8
0.35	1	0.125	0.222	0.964	1	1	0	7	8
0.5	0	0	0	0.958	0	0	0	8	8

Overall — all 5 subjects pooled

Every subject's matrix at the same thresholds, summed. Shows how the matcher behaves on the whole dataset, not just this subject.

threshold	precision	recall	F1	accuracy	retrieved	gold	TP	FP	FN
0.15	0.5	0.385	0.435	0.937	30	39	15	15	24
0.25	0.75	0.154	0.255	0.944	8	39	6	2	33
0.35	1	0.051	0.098	0.941	2	39	2	0	37
0.5	0	0	0	0.937	0	39	0	0	39

8. Limitations

1. **Testbed size** — 10 papers / 111 questions is a proof-of-concept scale, not yet production-scale.
2. **Lexical overlap dependency** — TF-IDF cannot detect matches with zero shared vocabulary; narrative-heavy scenarios dilute core terms.
3. **Threshold variance across disciplines** — Technical courses peak near $\tau \approx 0.25$; descriptive/applied subjects need lower thresholds ($\tau \approx 0.15$).
4. **OCR noise on notation** — Equations, matrices, and diagrams are prone to OCR degradation.
5. **Layout heuristics** — Regex cleaners are tuned to standard layouts and need adjustment for institutions with different header formats.

9. Future Scope

1. **Cross-department scaling** — Extend to Physical Sciences, Commerce, Management, Humanities, and Law.
2. **Dense semantic embeddings** — Integrate Sentence-BERT models (all-MiniLM-L6-v2, BGE-small-en) alongside TF-IDF for conceptual matching beyond lexical overlap.
3. **Entity & parameter masking (NER)** — Abstract away names, locations, and numeric values to isolate core theoretical concepts.
4. **Longitudinal trend analysis** — Cluster 5–10 years of data to reveal recurring themes and high-weightage modules.

5. **Multimodal diagram/formula recognition** — Incorporate LayoutLMv3, Nougat, or Surya OCR for equations and diagrams.
6. **LMS integration** — Deploy as a microservice API for Moodle, Canvas, or Google Classroom.

10. Conclusion

This project delivers a generalizable, end-to-end NLP system — paired with an interactive Streamlit dashboard — for extracting, segmenting, and analyzing cross-year exam questions to surface repeated topics across departments.

Techniques used: OCR, regex-based layout cleaning, state-machine segmentation, tokenization, stopword removal, WordNet lemmatization, N-gram TF-IDF, and cosine similarity.

Key results: Across 111 questions, the system reached a peak precision of **1.000** (τ = 0.35) and a balanced **F1-score of 0.435** (τ = 0.15), with the top-performing subject reaching an F1 of **0.640**.

Key takeaway: TF-IDF offers strong explainability and speed for detecting repeated exam questions, but pure lexical matching has ceilings — pairing it with transformer-based embeddings and narrative abstraction is the clear next step toward robust, cross-departmental deployment.

Github link : https://github.com/muhd-sinan-m/NLP-Project.git

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